

48mm OD Internally Wired Bar

Technical Datasheet - Generic Portal Reference

Document purpose: To provide a practical technical reference for internally wired lighting bars used in theatre and entertainment environments. This datasheet supports portal documentation, inspection planning and site review. It does not replace manufacturer instructions, electrical certification, design calculations, installation records or competent-person assessment.

Equipment type	48mm OD Internally Wired Bar
Portal category	Lifting Equipment / Lighting Bar
Typical classification	Load-bearing support bar with integrated electrical outlets
Document status	Generic reference template - edit for manufacturer and site details
Prepared for	Stage Electrics Mechanical Theatre Equipment Compliance Portal
Revision	Draft 1.0
Date	08 October 2026

Important limitation

This document is deliberately generic. The actual manufacturer, bar length, construction, wiring arrangement, outlet type, support spacing, SWL/WLL, electrical rating, inspection regime and site limitations must be confirmed before the datasheet is used as equipment-specific technical information.

1. Equipment overview

An internally wired bar is a 48mm outside-diameter lighting or suspension bar that contains electrical wiring, socket outlets or connection points within the bar. It may be fixed, suspended, flown, or incorporated into a wider lighting bar, grid or stage machinery arrangement depending on the installation.

- The bar should be assessed both as a load-bearing support and as electrical infrastructure.
- The bar, supports, suspension points, brackets, fixings, outlets, cables and supporting structure form one complete system.
- The bar should only be loaded within its confirmed SWL/WLL and any site-specific point-load or distributed-load limits.
- Electrical outlets, cable entries, end caps, glands and containment should be protected from impact, strain and mechanical damage.

- Only competent persons should alter, inspect, repair, load or modify the installation.

2. Equipment identification and technical details

Manufacturer / supplier	To be confirmed
Model / bar type	To be confirmed
Serial number / asset ID	To be confirmed
Location / installation reference	To be confirmed
Tube size	48mm outside diameter - confirm wall thickness and material
Bar length	To be confirmed
Safe Working Load / Working Load Limit	To be confirmed from design, manufacturer or competent-person information
Point-load / distributed-load limits	To be confirmed
Support spacing / suspension method	To be confirmed
Electrical outlet type and quantity	To be confirmed
Electrical supply / circuit details	To be confirmed by competent electrical person
Inspection interval	To be confirmed by written scheme, competent person or local inspection regime

3. Intended use

- Support luminaires and other approved production equipment within the known capacity of the bar and its supporting structure.
- Provide integrated electrical outlets or connection points for suitable production equipment where the installation has been designed for that use.
- Use only compatible clamps, couplers or brackets suitable for 48mm OD tube and the load being applied.
- Distribute loads evenly where possible and check concentrated loads against the bar design and support spacing.
- Maintain electrical and mechanical separation, strain relief and cable management so the bar is not used in a way that damages internal wiring.
- Do not use for lifting people or for unapproved rigging arrangements.

4. Key limitations and hazards

Hazard / limitation	Why it matters	Control / check
Unknown load rating	The bar may be overloaded if SWL/WLL, point loads or support spacing are not known.	Confirm ratings and limitations before loading.
Concentrated loads	Heavy loads at one point can create bending, local deformation or excessive stress at supports.	Check point-load limits and distribute loads where possible.
Electrical damage	Impact, drilling, crushed tube, poor glands or pulled cables may compromise the internal wiring.	Inspect outlets, glands, covers and cable routes; escalate defects to a competent electrical person.
Incompatible clamps	Unsuitable clamps may crush, slip on or damage the bar.	Use compatible clamps for 48mm OD tube and fit secondary suspension where required.
Poor support condition	Loose fixings, corroded supports or unsuitable structure can compromise the full load path.	Inspect brackets, suspension points, fixings and supporting structure.
Unauthorised modification	Drilling, welding, cutting or non-standard repair can affect strength and electrical safety.	Do not modify unless assessed and approved by competent persons.

5. Inspection and maintenance guidance

The following checks should be adapted to manufacturer instructions, electrical inspection requirements, the written scheme of examination and the local inspection regime. Any defect affecting load-bearing capacity, suspension, electrical safety or secure attachment should be escalated and the bar isolated or unloaded where necessary.

Area	Typical checks	Status	Notes
Identification and rating	Bar ID, location, SWL/WLL, loading limits and inspection status are known and legible where applicable.	☐ OK ☐ Defect	
Bar condition	Tube is straight and free from bending, dents, cracks, crushing, corrosion or signs of overload.	☐ OK ☐ Defect	

Area	Typical checks	Status	Notes
Supports and fixings	Brackets, clamps, drop rods, suspension wires, chains, bolts and anchors are secure and undamaged.	☐ OK ☐ Defect	
Load attachments	Luminaires and equipment are correctly clamped, not excessive for the bar, and fitted with secondary suspension where required.	☐ OK ☐ Defect	
Electrical outlets	Socket outlets, covers, end caps, glands and cable entries are intact, secure and undamaged.	☐ OK ☐ Defect	
Cable management	Cables are supported, not strained, trapped, abraded, crushed or causing snagging/trip hazards.	☐ OK ☐ Defect	
Modifications	No unauthorised holes, cuts, welded parts, added fixings, poor repairs or exposed wiring are present.	☐ OK ☐ Defect	
Operation / movement	Where flown or movable, travel is smooth and does not strain cables, supports or connected equipment.	☐ OK ☐ Defect	
Documentation	Inspection records, electrical test information, maintenance records and manufacturer/site information are available.	☐ OK ☐ Defect	

6. Pre-use checks

- Confirm the bar is suitable for the intended equipment and load arrangement.
- Check visible condition of the bar, support points, clamps and attached equipment before use.
- Check that luminaires and equipment are secure and that secondary suspensions are fitted where required.
- Check that socket outlets, plugs and cables appear intact and are not under strain.

- Do not add loads to a bar with unknown capacity or visible damage.
- Stop use and report defects if there is movement, distortion, electrical damage, loose fittings, overheating, burning smell or exposed wiring.

7. Regulatory and good-practice context

Framework	Relevance to internally wired bars
LOLER 1998	Relevant where the bar forms part of lifting equipment or is used to suspend loads at work. The load-bearing arrangement should be suitable, marked where appropriate and examined/inspected according to the risk and written scheme.
PUWER 1998	Applies as work equipment. The bar and its associated equipment must be suitable, maintained, inspected where deterioration could create risk, and used by trained people.
BS EN 17206	Relevant where the bar forms part of an entertainment lifting or moving system. Load-bearing components, movement, controls and prevention of unsafe failure should be considered.
ABTT Technical Standards	Supports theatre-industry good practice for installation, support, inspection, maintenance, documentation and competent operation in places of entertainment.
Electrical safety	Electrical inspection, testing, maintenance and repair should be carried out by competent persons following the site electrical safety regime and manufacturer information.

Competent-person note

The competent person should determine the inspection and thorough examination requirements for the specific installation, taking account of support arrangement, loading, environment, frequency of use, movement, electrical condition, modifications and consequences of failure.

8. Information to retain with this datasheet

- Manufacturer information, installation details and loading information where available.
- Electrical circuit schedules, test records and maintenance records where applicable.
- Support spacing, fixing type, structural information and any load calculations.
- Inspection reports, thorough examination reports where applicable and defect history.
- Records of repairs, modifications, replacement sockets, rewiring or relocation.
- Site-specific risk assessment and safe operating / loading procedure.

9. Portal upload details

Recommended PDF filename	internally_wired_bar_technical_datasheet.pdf
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Portal document title	48mm OD Internally Wired Bar Technical Datasheet
Related equipment ID	internally_wired_bar

10. Revision notes

Revision	Date	Change	By
1.0	08 Oct 2026	Initial generic portal datasheet.	Stage Electrics / ChatGPT support

Disclaimer: This datasheet is a guidance aid only. It must be reviewed and completed using actual manufacturer documentation, electrical information, installation records, design information and competent-person findings before being relied on as equipment-specific technical data.